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Review

Core, shapley value, nucleolus and nash bargaining solution: A Survey of recent developments and applications in operations management☆

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ABSTRACT

The core, Shapley value, nucleolus and Nash bargaining solution are the most widely used solution concepts in cooperative game theory. This paper surveys the literature on recent developments of these solution concepts and applications in operations management. Firstly, we briefly introduce the basic concepts and recent theoretical developments of the four solution concepts. Secondly, we review the applications in operations management in the past decade. Based on the problem properties, we classify the literature into deterministic and stochastic models, horizontal cooperation and vertical cooperation modes. Lastly, we discuss the future research directions with applications in operations management, which are hoped to contribute to the literature by enhancing the applications of cooperative game theory in operations management.

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1. Introduction

Cooperative game theory mainly studies the allocation of the payoff resulting from the cooperation of multiple players. When they cooperate and take joint action in a coalition, it often increases the total wealth and/or decreases the total cost of the coalition. Naturally, how to allocate the total payoff in a fair and stable manner is an interesting and important problem, which attracted many game scientists since the 1940s. Many solution concepts have been proposed to determine the allocation, and the core, Shapley value, nucleolus and Nash bargaining solution are the most widely used solution concepts in cooperative game theory.

In 1953, Gillies [1] introduced the concept of the core, the set of feasible allocations for which no coalition can gain more by leaving the grand coalition. Even though the core has been found useful in studying economic and management problems, it often does not provide a unique solution to the allocation problem. Also in 1953, Shapley [2] introduced the Shapley value which can be easily applied to compute the allocation by using a formula regardless of whether or not the core is empty. However, when the core is non-empty, the Shapley value may not be in the core, which means that the allocation in terms of the Shapley value may be unstable.

In 1969, Schmeidler [3] introduced the nucleolus solution concept, which is based on the principle of minimizing the unhappiness of the most unhappy coalition(s), of the second most unhappy coalition(s), etc. A coalition's unhappiness is characterized by the excess of the coalition, which is defined as the difference between the total payoff achieved by the members of the coalition and the sum of the players' payoffs that are allocated when they join the grand coalition. Schmeidler [3] also showed that if the core of a cooperative game is non-empty, then the nucleolus must be located in the core and thus assures the stability of the grand coalition. Unfortunately, it may not be easy to obtain the nucleolus solution, because we usually have to calculate this solution concept numerically by solving a series of linear programming (LP) problems in an iterative manner or by solving a large-scale LP problem. Moreover, it does not guarantee any of the monotonicity criteria.

When two players cooperate to generate a surplus, how to share the surplus is always accompanied with a bargaining problem. Based on some appealing axioms, Nash (1950) proposed a solution to solve such cooperative bargaining problem, which is called the Nash bargaining solution later. However, Nash bargaining approach is only suitable for two players' cooperative problem.

This paper aims to provide a systematic review of the recent developments and applications in operations management of the core, Shapley value, nucleolus and Nash bargaining solution. Nagarajan and Sosic [4] conducted a survey of the applications of cooperative game theory in supply chain management, focusing on the problem of how to allocate the profit for the stability. The first

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part of this paper mainly discusses how to use the cooperative bargaining model to divide the total pie resulting from cooperation, and the second part investigates the coalition formation among the supply chain partners. Dror and Hartman [5] conducted a comprehensive survey of cooperative inventory games, focusing on the deterministic games such as EOQ-like policies with joint replenishment, and the stochastic games such as newsvendor-like centralization games and the extensions. Different from the above two review papers, we focus on the recent developments and applications in operations management in the past decade, which complement the survey on this topic.

The remainder of this paper is organized as follows. In Section 2, we introduce the basic concepts and recent theoretical developments of the core, Shapley value, nucleolus and Nash bargaining solution. The applications of these solution concepts in operations management in the past decade are surveyed in Section 3. This paper ends with the conclusion and some potential research directions.

2. Core, shapley value, nucleolus and nash bargaining solution with recent developments

An $n-$player cooperative game in characteristic form is defined by (N, v) (or (N, c)), where $N = \{1, 2, \cdots n\}$ denotes the set of all players, and $v(S)$ (or $c(S)$) is the characteristic function for the subset (i.e., coalition) $S \subseteq N$. And in the game (N, v) (or (N, c)), x_i is often used to represent the payoff to the ith player. For a game (N, v), it is said to be convex if $v(S \cup T) + v(S \cup T) \geq v(S) + v(T)$, for any $S, T \subset N$, whereas a cost game (N, c) is concave if $c(S \cup T) + c(S \cup T) \leq c(S) + c(T)$, for any $S, T \subset N$. Later, for ease of exposition, we just introduce the solution concepts for (N, v). A payoff vector $\mathbf{x} = \{x_1, x_2, \cdots x_n\}$ is called a pre-imputation if $\sum_{i \in N} x_i = v(N)$, and is defined as an imputation if $\sum_{i \in N} x_i = v(N)$ and $x_i \geq v(\{i\})$.

2.1. Core

The core of a cooperative game is the set of all imputations which ensure the stability of the grand coalition, that is,

$$C(N, v) = \{\mathbf{x} = \{x_1, x_2, \cdots x_n\} \mid \sum_{i \in N} x_i = v(N),$$
$$\text{and } \sum_{i \in S} x_i \geq v(S) \text{ for all } S \subset N\},$$

where $C(N, v)$ represents the core of the game.

The nonemptiness of the core can guarantee that there is at least an allocation such that no any player or coalition has the incentive to leave the grand coalition. Thus, for a cooperative game, determining whether the core of the game is nonempty is of significant importance. The classical method to check the nonemptiness of the core is the Bondareva-Shapley theorem, which states that the core of a cooperative game is non-empty if and only if the game is balanced (Bondareva 1968 and Shapley 1967 [6,7]). A cooperative game is balanced if for every balanced set $\delta : 2^N \backslash \{\varnothing\} \longrightarrow [0, 1]$ such that $\forall i \in N$, $\sum_{S \in 2^N : i \in S} \delta(S) = 1$, the following condition holds:

$$\sum_{S \in 2^N \backslash \{\varnothing\}} \delta(S) v(S) \leq v(N).$$

And a game (N, v) is called to be totally balanced if all subgames (S, v_S), $\emptyset \neq S \subseteq N$ are balanced. The Bondareva-Shapley theorem implies that a game is totally balanced if and only if any subgame has a non-empty core.

The Bondareva-Shapley theorem requires to check every balanced set, which is usually very difficult. Thus, many scholars tried to find a simple method to check the nonemptiness for a certain class of cooperative games in practice.

Motivated by supply chain collaborations in practice, Meca and Sosic [8] introduced a class of cost-coalitional problems with decreasingly monotonic coalitional costs, that is, $c_i^S \leq c_i^T$ for all $i \in T$ and $T \subset S$, and derived the condition for guaranteeing the stability of the grand coalition in terms of the core.

For a zero-normalized nonnegative game, Abe [9] showed that the core is nonempty if and only if the game can be decomposed into some N -monotonic veto-controlled games. For multi-player cooperative game, Gutierrez et al. [10] investigated the core stability and strong equilibrium and found that if the strong equilibrium model is feasible, its optimal cost solution is just the optimal core cost allocation.

A newsvendor game is a cooperative game where the newsvendor players collaborate by pooling their inventory and sharing the corresponding total cost. The core of such game is proved to be nonempty, but the allocation in the core may not guarantee the unhindered formation of a coalition. Chen et al[11]. proposed the concept of population monotonic allocation scheme (PMAS), which requires that (i) $\sum_{i \in S} l_{i,S} = c(S)$, for all $S \subseteq N$, (ii) $l_{i,S} \leq l_{i,T}$, for all $i \in S \subseteq T \subseteq N$, where $l_{i,S}$ is the amount of cost allocated to player i in coalition S. Under some mild conditions, the authors showed that the PMAS exists in newsvendor games and the traditional dual-based allocation scheme is actually a PMAS. Motivated by the cooperative game associated with the joint replenishment model, He et al. [12] studied maximizing a separable concave function over a polymatroid and explored the submodularity of the optimal objective value in the parameters of the objective function. Applying the research results, they showed that the joint replenishment cooperative game is submodular if the joint setup cost is a normalized and non-decreasing submodular function.

Lozano et al. [13] investigated a set-valued DEA (Data Envelopment Analysis) production game where a number of decision-makers cooperate by pooling their resources and sharing their technologies in the production process, and then divide the final production. For such a multi-commodity game, they extended two core concepts, i.e., the preference core and the dominance core, based on different excess functions to measure the dissatisfaction of the coalitions. For some cooperative games, Anily [14] designed a cost allocation contract among the cooperative firms. Specifically, for the regular game and the centralizing aggregation game, the author derived the sufficient conditions for the total balancedness and a full characterization for the nonempty core.

Motivated by the belief that risk aversion promotes cooperation, Yang and Li [15] extended the traditional core to the contending core and investigated the impact of risk aversion on the formation of coalitions. Using the Conditional Value at Risk (CVaR) as the risk measure, the research shows that the more they worry about the worst cases, the more likely they will band together.

Based on the well-known law of diminishing returns in economics, Leng et al. [16] proposed the almost diminishing marginal contributions (ADMC) games, that is

$$v(i) \leq v(T \cup \{i\}) - v(T) \leq v(S \cup \{i\}) - v(S),$$
$$\forall i \in N, \forall S \subseteq T \subseteq N \backslash \{i\}, \text{ and } S \neq \varnothing,$$

and found that the core is nonempty if and only if

$$\sum_{j \in N} v(N \backslash \{j\}) \leq (n-1) v(N).$$

To illustrate the applications of this result, they analyzed three real games in operations management: a code-sharing game, a group-buying game, and a scheduling profit game.

If the core of a cooperative game is empty, then no allocation scheme can ensure the stability of the grand coalition, because at least one coalition can gain more by leaving the grand coalition.

Under such background, assuming that each coalition needs to pay a penalty of ε for leaving the grand coalition, scientists have raised the concept of ε-core to characterize the set of allocation schemes guaranteeing the grand coalition's stability. When the value of the penalty ε is minimized subjected to non-emptiness of the ε -core, the corresponding ε-core is called the least core (Maschler et al. 1979[17]), which can be treated as a generalization of the core. According to Maschler et al. [17], the least core of a coalitional game includes all the pre-imputations that are optimal solutions to the following linear programming problem:

$$\varepsilon^* = \min\left\{\varepsilon \mid \sum_{i\in N} x_i = v(N); v(S) - \sum_{i\in S} x_i \le \varepsilon, \forall S \subset N, \text{and } S \ne \varnothing\right\},$$

where ε^* is the least core value.

Because every supermodular cost game has an empty core, Shulz and Uhan [18] proposed a 3-approximation algorithm for computing the least core value of supermodular cost cooperative games, and a 2-approximate algorithm for a cost allocation in least core. These algorithms computing the least core value and a cost allocation in least core can be extended to the submodular profit cooperative games. The research is applied to analyze the classical scheduling games.

When the core is empty, then at least a player/coalition can get more payoff by leaving the grand coalition, and thus the grand coalition is not stable. Liu et al. [19] used the simultaneous penalization and subsidization strategy to stabilize the grand coalition, which charges a penalty from leaving players and provides a subsidy to the players staying in the coalition. Such reward and penalty mechanism is also applied to analyze the water resource environmental problem [20].

When the core of an ADMC games is empty, Leng et al. [16] showed that the least core value satisfies $\varepsilon^* \ge [\ v(N) - \sum_{j\in N}(v(N) - v(N\backslash\{j\}))]/n$, and obtained a closed-form solution for the least core value under a certain condition. The results are also applied to analyze the classical prize donation game.

2.2. Shapley value

The Shapley value is a direct approach to allocate the total payoff among all players when they form a grand coalition to cooperate. According to the Shapley value, the amount that the player i can obtain is

$$\phi_i(N, v) = \sum_{S\subseteq N\backslash\{i\}} \frac{|S|!(n-|S|-1)!}{n!}(v(S\cup\{i\}) - v(S)),$$

where n is the total number of players, $|S|$ is the cardinality of the coalition S. The Shapley value allocation is relatively fair in the sense of players' contribution, because each player's allocation is developed on the basis of his contributions to all possible coalitions. And the Shapley value has an explicit allocation formula. Thus, many researchers and practitioners would like to apply it to divide the grand payoff. The Shapley value satisfies both aggregate and coalitional monotonicity.

The Shapley value $\phi_i(v)$ can be rewritten as

$$\phi_i(N, v) = \sum_{S\in 2^N:\ S\ni i} \Delta_v(S)/s,$$

where $\Delta_v(S) = \sum_{T\subseteq S}(-1)^{|S\backslash T|}v(T)$ is the Harsanyi dividend. Shapley (1953) also proposed the weighted Shapley value as

$$\widehat{\phi}_i(N, v) = \sum_{S\in 2^N:\ S\ni i} \frac{w_i}{\sum_{j\in S} w_j}\Delta_v(S),$$

where w_i is the weight of player i. Casajus [21] re-investigated all the weighted Shapley value and found that the Shapley value is the unique positively weighted Shapley value that satisfies the efficiency and the null player out property.

Beal et al. [22] extended the classic Shapley value to the proportional Shapley value, by incorporating different players' standalone worth, as

$$\overline{\phi}_i(N, v) = \sum_{S\in 2^N:\ S\ni i} \frac{v(\{i\})}{\sum_{j\in S} v(\{j\})}\Delta_v(S).$$

Because the weights are endogenous, this extension may be more intuitive in various applications of cooperative games.

Skibski et al. [23] investigated the coalitional game with externalities which can be characterized as a discrete-time stochastic process, and used the probability theory to study the Shapley's coalition formation process. Besner [24] proposed the weighted Shapley hierarchy levels values for the cooperative games with level structure, and showed that such extension can guarantee two well-known axiomatizations characterized by efficiency and weighted balanced contributions, and by weighted standardness for two-player games and consistency. Gusev [25] introduced the vertex cover game, which exists widely in transport networks, society, computer networks, business processes, etc., and investigated the calculation of the Shapley value of such game and the application in transport networks.

The computation of the Shapley value is exponentially difficult because it requires considering the contributions to all subsets. Especially, when the size of the game is large, the computation is very hard. Based on the stochastic approximation of deterministic games and sampling theory, Benati et al[26]. proposed a stochastic approach to calculate the estimates of the Shapley values. Specifically, the approximate Shapley value is

$$\widetilde{\phi}_i(N, v) = \frac{v(N)}{n} + \frac{1}{\tau}\left[\sum_{S\in\mathbf{S}, i\in S}\left(\frac{n-1}{s}\right)v(S) - \sum_{S\in\mathbf{S}, i\notin S}\left(\frac{n-1}{n-s}\right)v(S)\right]$$

where $\mathbf{S} = (S_1, S_2, \cdots, S_\tau)$ is a random sample of the family of all the coalitions in this game. Some well-known games are applied to testify the efficiency of such calculation method. Hou et al. [27] generalized the well-known class of airport game to an airport cost pooling game, and applied a decomposition method to calculate the Shapley value of such class of games.

The Shapley value can give rise to a unique allocation for any cooperative game, but it cannot guarantee the stability of the allocation. For an ADMC game with a non-empty core, Leng et al. [16] showed that if

$$\sum_{i\in S}[v(i) + (n-1)(v(N) - v(N\backslash\{i\}))] \ge nv(S)$$

the Shapley value lies in the core which guarantees the stability of the grand coalition.

The core concept is developed based on the stability, while the Shapley value is designed for fairness in the sense of contribution. However, the Shapley value may not lie in the core and the core allocation may ignore the fairness of contribution. Nguyen [28] proposed a new solution concept called the fairest core which takes both the stability and fairness into consideration, and analyzed the attractive properties.

2.3. Nucleolus

The nucleolus solution is an important concept in cooperative game theory, which is defined as a pre-imputation $\mathbf{x} = \{x_1, x_2, \cdots, x_n\}$ in a cooperative game (N, v) such that the excess $e_S(\mathbf{x}) = v(S) - \sum_{i\in S} x_i$ of any possible coalition S cannot be reduced without increasing any other greater excess.

Formally, let $\theta(\mathbf{x})$ be a vector in R^{2^n} with the numbers $e_S(\mathbf{x})$ ($S \subseteq N$, $S \neq \varnothing$) as its coordinates, arranged according to their magnitude in decreasing order. The lexicographic order on R^{2^n} induces a quasi-order among pre-imputations. The nucleolus of the game is defined as the set of points which are the first of all pre-imputations according to the quasi-order law, that is,

$$\mathcal{N} = \{\mathbf{x} \in \mathbf{Y} |\ \theta(\mathbf{x}) \preceq \theta(\mathbf{y}), \text{ for all } \mathbf{y} \in \mathbf{Y}\}$$

where $\mathbf{Y}$ is the set of all pre-imputations.

Although the nucleolus allocation approach has some desirable properties (for example, it is always located in the core if the core is non-empty), it is very difficult to calculate. So many researchers worked hard on the algorithms. Puerto and Perea [29] investigated how to calculate the nucleolus of any n-person cooperative game by a single linear programming. Based on sampling the coalitions space, Perea and Puerto [30] found a heuristic procedure for searching the nucleolus, and their experiments shows that the allocations resulting from the heuristic procedure are very close to the true nucleolus.

Nguyen and Thomas [31] proposed a method based on nested linear programming to compute the nucleolus of the large cooperative games, including some games with more than 50 players. Moreover, they studied some technical issues such as how to deal with multiple optimal solutions and tight constraint sets, and proposed a new nested linear programming formulation which is smaller in terms of the linear programming size.

For ADMC games, Leng et al. [16] showed that a player's allocation in terms of the nucleolus solution is only dependent on the players' marginal contributions to the grand coalition. It is worth noting that Okada [32] also got a similar result. Specifically, for a three-person cooperative game with different levels of efficiency, the author investigated the stationary equilibria and showed that the players' expected equilibrium payoffs also depends mainly on the marginal contributions of players to the grand coalition.

Schaarsberg et al. [33] investigated the nucleolus of the maximum cooperative purchasing (MCP) games, where the unit price depends on the largest order quantity within a group of players, and obtained an explicit alternative mathematical characterization. The research also shows that the core is nonempty if the unit price is decreasing. Moreover, the authors found an explicit expression for the Shapley value by the decomposition of an MCP-game into unanimity games.

Hu et al. [34] studied the nucleolus of the nested cost-sharing problems where some public facilities are shared by many agents having different needs, and found that the nucleolus is the only Right-endpoint Subtraction (RS) bilaterally consistent (or RS conversely consistent) rule satisfying equal treatment of equals and last-agent cost additivity. To guarantee the stability of cooperative games with externalities, Tripathi and Amit [35] proposed a new allocation method called Equivalence nucleolus, and showed that such solution concept is unique and always nonempty.

2.4. Nash bargaining solution

Nash bargaining is an approach in which two players decide how to share a surplus when they cooperate to obtain. Nash bargaining solution [36] is the unique solution that satisfies the axioms of symmetry, efficiency, scale invariance and independence of irrelevant alternatives, which is exactly the point (x^*, y^*) that maximizes the following Nash product

$$(x^*, y^*) = \arg\max_{(x,\ y)\in F} (u(x) - u_0)(v(y) - v_0),$$

where $u(x)$ and $v(y)$ are the utility functions of player 1 and player 2, u_0 and v_0 are the status quo utilities, and F is the feasible set.

Nash bargaining solution implicitly assumes that two players have the same bargaining power. In reality, two players may have different bargaining power, and suppose β and $(1-\beta)$ are the relative bargaining power of player 1 and player 2 respectively, then the generalized Nash bargaining solution (Roth, 1979[37]) is obtained by

$$(\tilde{x}^*, \tilde{y}^*) = \arg\max_{(x,\ y)\in F} (u(x) - u_0)^{\beta} (v(y) - v_0)^{1-\beta}.$$

Lippman and McCardle [38] proposed the embedded Nash Bargaining approach to model the joint decision making and investigated its existence, uniqueness, and comparative statics. Moreover, they showed that a decision maker's payoff decreases (increases) in both his risk aversion and impatience (in his opponent's risk aversion and impatience), when the disagreement payoff is random.

A virtual bargaining equilibrium is a feasible agreement that maximizes the following product

$$(\sigma_1^V, \sigma_2^V) = \arg\max_{(\sigma_1^F, \sigma_2^F)\in F} \prod_{i=1}^{2} (w_i(\sigma_1^F, \sigma_2^F) - w_i^m)$$

where $w_i(\sigma_1^F, \sigma_2^F) \equiv \min\{u_i(\sigma_1^F, \sigma_2^F), \sup_{\sigma_2 \in R_2(\sigma_1^F)} u_i(\sigma_1^F, \sigma_2)\}$ is the worst payoff of an agreement (σ_1^F, σ_2^F). Melkonyan et al. [39] used the virtual bargaining scheme to provide a novel explanation of collusive behavior that distinguishes between outcomes in Bertrand and Cournot competition. Virtual bargaining contains elements of both noncooperative and cooperative reasoning which sustains collusive outcomes without third-party enforcement. The research also reveals why collusion in price competition can be sustained without communication and is hard to detect.

Qin et al. [40] investigated the implementation of Nash Bargaining and showed that the solutions can be implemented as the unique subgame equilibrium payoff allocations. Lozano et al. [41] extended the bargaining approach for computing DEA targets, and showed how the Nash bargaining, the lexicographic Kalai–Smorodinsky, the weighted lexicographic egalitarian and the normalized utilitarian solutions give rise to corresponding DEA bargaining models.

3. Recent applications in operations management

In this section, we introduce the recent applications of the core, Shapley value, nucleolus and Nash bargaining solution in deterministic operational models and stochastic operational models.

3.1. Deterministic operational models

According to the structure of the cooperation model, we review the horizontal cooperation and vertical cooperation models respectively.

3.1.1. Horizontal cooperation

Horizontal cooperation means that the firms are operating at the same level in the market and cooperate to form mutually beneficial relationships. One class is assuming that the external environment is exogenous and there is no interaction with external firm, and we denote it by H-Type 1 (i.e., Horizontal Type 1). The other class, H-Type 2, is that the firms' cooperation will interact with the external firm, i.e., the characteristic functions depend on the interaction. Fig. 1 lists two horizontal cooperation modes.

Horizontal Type 1

In reality, many firms may pool their facilities, inventories, orders, etc., cooperating to achieve better performance. The crucial problem is how to determine the characteristic functions based on the practical problem.

Guo et al. [42] developed a cooperative game to explore how to divide the reduced number of staff fairly among the call centers that pooled their capacities. The research shows that such

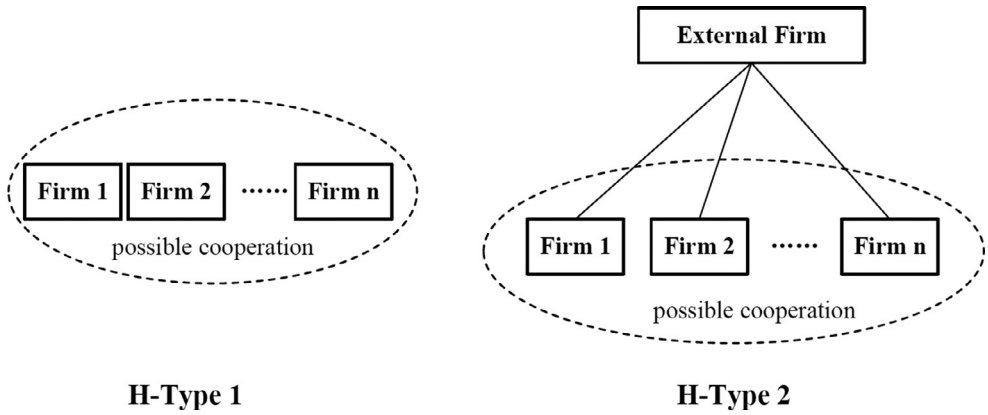

Fig. 1. Horizontal cooperation modes.

game is convex with a non-empty core, and provides a neat Shapley value_characterized staff-allocation rule which is located in the core of the game. Because in practice, some retailers exchange their excess space to improve their performance (e.g., Waitrose and Boots/Welcome Break, Tim Hortons and Cold Stone Creamery), Leng et al. [43] applied the generalized Nash bargaining scheme to divide the system-wide surplus for the two retailers, and used the weighted Shapley value to allocate the surplus for more retailers. Jin et al. [44] studied the problem of sharing container storage space among multiple companies in a port. Based on the linear programming model, the authors proposed an ideal cost allocation which lies in the core of the cooperative game.

Fiestras-Janeiro et al. [45] examined the cooperative game of a group of agents that all face a continuous review inventory problem with a limited capacity warehouse. When there is no holding cost, they found that the grand coalition is profitable and proposed a natural easy allocation among agents which is also stable. Finally, they compared the allocation method with the Shapley value allocation method, and the nucleolus allocation method. For expensive, low-demand spare parts of the high-tech machines, the independent firms often collaborate by pooling their inventories via free transshipments. Karsten et al. [46] used the cooperative game theory to investigate the stability of such a pooling game, and examined the issue of how to fairly allocate the collective holding and downtime costs among the firms. Specifically, they studied two settings: one is that each party maintains a predetermined stocking level and the other is that based-stock levels are optimized. Although the research is motivated by spare parts applications, the results are also applicable to other pooled resource systems similar to an Erlang loss system. Estevez-Fernandez and Reijnierse [47] investigated the cost sharing problem of a general service by explicitly taking the generated revenues into consideration. For such a cost-revenue sharing problem, they proposed a cost-revenue cooperative game where the value of a coalition is defined as the maximum net revenues that the coalition may obtain by means of cooperation, and investigated the core of the game. The research shows that a cost-revenue cooperative game has a nonempty core if and only if the dual game of the cost game has a large core.

Kimms and Kozeletskyi [48] formalized the horizontal cooperation of traveling salesmen as a cooperative game, and used the core concept to explore its cost allocation scheme. Based on mathematical programming techniques, they developed a core computation algorithm to find a core element when the core is nonempty, or a least-core element in the case of an empty core. Kimms and Kozeletskyi [49] proposed a concept based on the core to tackle the cooperative game problem with non-transferable and transferable utilities, and obtained a mathematical programming to solve such a multi-objective optimization problem. Osicka et al[50]. applied the cooperative game-theoretic framework to investigate several variants of the location-routing problem, and found that when the facility opening cost is significantly larger than the routing cost, the core of the game must be nonempty. Logistics collaboration in forestry and oil operations will greatly reduce the operations cost, Guajardo and Rõnnqvist [51] used the mixed integer linear programming to investigate the core stability and strong equilibrium of collaborative logistics. The research also implies that the marginal contribution to savings significantly decreases as the size increases.

Bernstein et al. [52] modeled the knowledge exchange as a cooperative game among suppliers where all suppliers can achieve the cost reduction due to the cooperation. They identified the cost allocation rules based on the core of the knowledge sharing game, and proposed the Altruistic allocation rule and Tute allocation rule, and compared the various cost allocation rules. Lu et al. [53] used the cooperative game theory to study how different firms can cooperate in the recycling and pricing decisions, and designed the allocation schemes of the maximized total profit to encourage cooperation among all firms.

Hall and Liu [54] investigated a capacity allocation problem in which a number of manufacturers order production capacity and the orders occur a value and time-dependent scheduling cost. If there is no initial sequence of orders, they examined the exact condition for the cooperative game to be balanced. Mitridati [55] explored how to design a community-based market mechanism when prosumers can pool their heat and electricity production and consumption, and examined the allocation schemes based on the core, Shapley value and nucleolus, etc., for such cooperative energy system. Zheng et al. [56] applied the multi-choice goal programming cooperative game approach to help design an incentive scheme for producer responsibility organization of waste tires, which is more suitable than the Shapley value method in this situation.

Horizontal Type 2

For a manufacturer serving a number of retail stores who all face deterministic demands in a finite planning horizon, he needs to build the production capacity, and then to produce and distribute products to the retail stores. Because the retail stores are different independent firms, how to divide the total cost among the retail stores is an important problem. Chen and Chen [57] showed that under various conditions, there is a fair core allocation of costs among the retail stores, and provided the mechanism to calculate a fair cost allocation. Multiple retailers form a coalition by placing joint orders to a single supplier in order to reduce ordering cost, Chen and Zhang [58] showed that the core of such cooperative game is nonempty when the inventory holding cost and backlogging cost are both linear functions. Moreover,

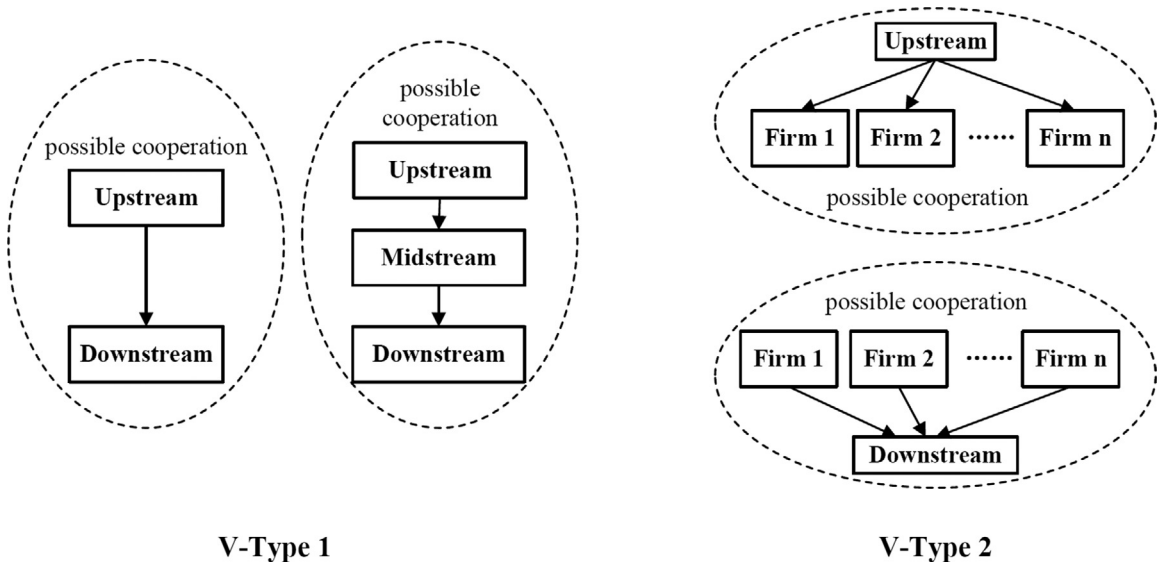

Fig. 2. Vertical cooperation modes.

when all retailers have the same cost parameters, a core allocation can be computed in polynomial time. Hezarkhani et al[59]. investigated the collaborative replenishment in the presence of intermediaries and examined the conditions for the concavity of associated cooperative games. The research shows that the submodular cost functions cannot imply the concavity of the game and the Shapley value allocation can coordinate the supply chain even under asymmetric information.

In an assignment market, one seller owns several indivisible heterogeneous goods, and a number of buyers want to buy up to a given capacity and they can cooperate. The competitive equilibrium prices can not guarantee the core allocations, Robles and Nuñez [60] showed that the convexity of the associated coalitional game can guarantee that the buyers-optimal core allocation is competitive.

Nagarajan et al. [61] studied the dynamic stable supplier coalitions in an assembly system where multiple suppliers sell complementary components to a downstream assembler, and investigated the structure of the dynamic stable supplier formation.

Cloud manufacturing has made the mass manufacturing resources accessible to more clients, and clients may cooperate by exchanging and reutilizating their booked resources. Chen et al. [62] formulated such a problem as a cooperative game, and proposed a core cost saving allocation method which will benefit each client.

3.1.2. Vertical cooperation

Two often-used vertical cooperation modes in operations management are listed in Fig. 2. The first class is purely linear (Vertical Type 1, V-Type 1), and the other class involves one-to-multiple cooperation (V-Type 2).

Vertical Type 1

It is very common that some firms may cooperate by taking joint actions to achieve more profits. Guo et al. [63] used the nucleolus and the Shapley value to study the cooperative game where an issuer, an acquirer, and a merchant bargain over the interchange fee rate and the merchant discount rate in a credit card network. They derived both the nucleolus and the Shapley value-characterized rates, as well as globally-optimal rate for the grand coalition. The research also implies that to guarantee the stability of the grand coalition, the merchant should delegate the credit card business only to the issuer and the acquire with sufficiently low operation cost. Qing et al. [64] investigated a capacity-allocation problem with downstream competition and bargaining. The research shows that the firms can benefit from each other's production cost improvement and the higher bargaining power can backfire on the downstream firm.

Zheng, Liu et al. [65] used the cooperative and noncooperative approaches to explore a three-echelon closed-loop supple chain in the presence of fairness concerns. They provided three mechanisms which are respectively based on the nucleolus solution, the Shapley value and the equal satisfaction to divide the surplus profit, and studied the impact of fairness concern on the profit allocation under different mechanisms. Zheng, Li et al. [66] incorporated the fairness concern in the coordination of a closed-loop supply chain and proposed a weighted Shapley value to allocate the centralized profit. The results showed that the weighted Shapley value conditionally outperforms the classic Shapley value in the presence of fairness concerns. Chu et al. [67] used the Nash bargaing approach with the satisfaction degree as the payoffs to investigate the fixed cost allocation problem among decision-making units, and obtained a fixed cost allocation which is a bargaining equilibrium of the two unions.

Vertical Type 2

Aghadadashli et al. [68] applied the Nash bargaining theory to re-investigate the bargaining solution between an upstream and downstream firms, where they bargain over a linear input price with unobservable contracts. The research shows that the profit sharing rule depends on the elasticity of the demand.

Leng and Parlar [69] applied the cooperative game theory to investigate the transfer pricing decisions in a multidivisional firm where the upstream and downstream divisions need to negotiate the transfer price. The appropriate transfer price implies a fair allocation of the cooperative profit. The research shows that the game is superadditive and convex with a non-empty core. Based on the Shapley value approach, the authors proposed a unique, fair allocation scheme which can assure the stability of the grand coalition, and the analytical transfer-pricing decisions are put forward.

In today's social network, cooperative advertisement is a key factor to improve advertising effectiveness for companies. Hafezalkotob et al. [70] investigated such cooperative advertising problems, and used the Shapley value and core center to allocate the total benefit resulting from the cooperation. The research also implies that the fair allocation is very important to the companies' satisfaction and long-term cooperation.

Table 1
Deterministic Models.

Article	Topic	Model Properties		Solution Concepts			
		Single/Multi Period	Vertical/Horizontal	Core	Shapley Value	Nucleolus	Nash Bargaining
Aghadadashli et al. [68]	profit sharing	single	verti.				•
An et al. [71]	resource sharing	multi	horiz.		•		
Bernstein et al. [52]	knowledge sharing	multi	horiz.	•	•		
Chen et al. (2016)	resource sharing	single	horiz.	•			
Chen et al. (2013)	capacity sharing	multi	horiz.	•			
Chen et al. [11]	inventory cooperation	single	horiz.	•			
Chu et al. [18]	fixed cost allocation	single	verti.				•
Estevez-Fernandez et al.(2014)	cost sharing problem	single	horiz	•			
Fiestras-Janeiro et al. [45]	inventory sharing	multi	horiz.	•	•	•	
Guajardo et al. (2015)	collaborative logistics	single	horiz.	•			
Guo et al. [63]	credit card cooperation	single	horiz.	•	•	•	•
Guo et al. [42]	capacity pooling	single	horiz.	•	•		
Hafezalkotob et al. [70]	cooperative advertising	single	horiz.	•	•		
Hall et al. (2016)	capacity allocation	single	horiz.	•			
Hezarkhani et al. [59]	collaborative replenishment	multi	horiz.	•	•		
Jin et al. [44]	storage space sharing	single	horiz.	•	•		
Karsten et al. (2012)	inventory pooling	single	horiz.	•			
Kimms et al. (2016)	cooperative travelling	single	horiz.	•			
Kimms et al. (2017)	multi-objective travelling	single	horiz.	•			
Leng et al. (2012)	transfer pricing problem	single	verti.	•	•		
Leng et al. [43]	space-exchange problem	single	horiz.	•	•		•
Lu et al. [53]	recycling cooperation	single	horiz.	•			
Mitridati et al. [55]	energy allocation	single	horiz.	•	•	•	
Nagarajan et al. [61]	component cooperation	multi	horiz.	•	•		
Osicka et al. [50]	location-routing problem	single	horiz.	•	•	•	
Qing et al. [64]	capacity sharing	single	verti.				•
Robles et al. (2019)	assignment	single	verti.	•			
Zheng, Li et al. [66]	closed-loop supply chain	single	verti.	•	•		
Zheng, Liu et al. [65]	closed-loop supply chain	single	verti.	•	•	•	

Note: verti.–vertical, horiz.–horizontal.

In reality, the allocation of resource sharing is a dynamic problem, which relies on not only the multiple players themselves but also the stages. An et al[71]. integrated the network data envelopment analysis and Shapley value method to explore the resource sharing and payoff allocation problem in a three-stage system, and the research shows that the game among stages is superadditive.

Table 1 summarizes the research topics, model properties and solution concepts involved of the above reviewed papers.

3.2. Stochastic operational models

3.2.1. Horizontal cooperation

Horizontal Type 1

In the service industry, the demand is often stochastic and the resourcing pooling may be an efficient way to handle uncertainty. For example, several independent service providers such as hospitals may cooperate by pooling their resource and the customer streams into a joint service system, Karsten et al[72]. modeled such service systems as Erlang delay systems, used the cooperative game theory to divide the collective costs of the pooled system among the servers, and explored the sufficient condition to guarantee a core allocation. In reality, several health care regions may decide to pool their ambulances and cooperate by reallocating the available resources to those who need them most. Motivated by these practices, Schlicher et al. [73] formulated a probabilistic resource pooling game and investigated the cost savings allocation. The research showed that the core of such game is nonempty and proposes an intuitive class of allocation rules in the core.

Cooperation among servers may be one of the best approaches to reduce the overall congestion in a service system. Consecutive servers pool their resources, then the serve subsystems merge into a single combined server with an aggregated service demand and a greater service rate. Defining the steady-state mean total number of customers in the subsystem as the cost of such coalition, Bendel and Haviv [74] developed a cooperative game model and found that the core of the game is nonempty. The research also investigated the nucleolus, the Shapley value and the Banzhaf value, and found the three values are the same when the servers have equal mean service demands.

For a cooperative service system in which a number of servers pool their capacity to serve the customers, Anily and Haviv [75] investigated the production and service systems consisting of two types of lines, M/M/1 lines and lines without buffers. To reduce the cost of the system, the supervisors may cooperate by applying line-balancing techniques, possibly combined with strategic outsourcing or capacity reduction practices. The research shows that the cooperative games are market games and the core of the games and the subgames are both non-empty. Moreover, the research derives a core cost allocation for each game. In reality, different independent operators of queueing systems may cooperate by allowing capacity transfer among them to generate a win-win situation. And they will incur two types of costs, one is the congestion cost in the queueing system, and the other is capacity transfer cost. Zeng et al. [76] investigated the cost allocation problem under the M/M/1 and M/M/s queueing systems.

Because the capacities in a production system or service system will occupy a large percentage of fixed cost and some capacities remain low utilization. Under such background, capacity sharing among firms will save a lot of cost and improve the efficiency. Hu et al. [77] investigated the operations of an airline alliance in which independent firms can cooperate to market and operate codeshare and interline itineraries, and derived a revenue sharing such that no airline coalition has enough incentive to leave from the grand coalition. Airline alliance allows the airlines to extend their networks and thus increase the number of destinations they can access. For such alliance, how to share fairly the revenue resulting from selling a ticket for a product among the in-

dividual airlines is critically important for the long-term stability. Kimms and Çetiner [78] addressed the problem using the cooperative game theory and found that the core of the game is nonempty. Based on the concept of the nucleolus, the authors developed a revenue proportion scheme and proposed the revenue sharing heuristic. Moreover, Çetiner and Kimms [79] incorporated the airline's self behavior and proposed a method to assess the fairness of the revenue sharing mechanism which is based on the transfer of dual prices. On the other hand, Firms such as airplanes and power plants, etc. often face high cost due to downtime of their technical systems and holding spare parts. If the firms can cooperate by pooling common spare parts, the cost can be reduced. Then a natural question is how to divide the resulting cost or benefits in a fair manner. Karsten and Basten [80] proposed to allocate the total costs according to the players' demand rate. And the research shows that such allocation method satisfies many appealing properties. For example, it makes all subgroups better off, it is easy to implement in practice and can stimulate growth of the pooling, etc.

When a number of job owners share a single machine and the machine is unavailable for a period of time, the jobs will need to be rescheduled. Liu et al[81]. modelled the rescheduling problem as a cooperative sequencing game, and derived a simple closed-form core allocation of the total cost and the Shapley value of the game. The authors also investigated the likelihood of the Shapley value locating in the core.

Timmer et al. [82] investigated the stable cost allocations of the joint costs when the collaborative companies face Poisson demand and review the inventory continuously. The research shows that the Shapley value and the distribution rule are stable cost allocations. Li et al. [83] investigated the inventory game with permissible delay in payment and showed that the core of such game is nonempty. Moreover, the research proposes a core allocation rule which guarantees the farsighted stability of the grand coalition. Lai et al. [84] investigated how to allocate the cost for the cooperative inventory consolidation programs, and proposed a proportional cost allocation rule for the time-based policy and the quantity-based policy. For electricity systems, Hagspiel [85] investigated the reliability and the role of interdependent suppliers. Based on the concept of Shapley value, the research provides a unique consistent reliability allocation rule, which is also applied to the case of wind power contributing to the reliability of Germany's electricity system.

Horizontal Type 2

Facing a stochastic market, excessive inventory or stocking less will both result in poor performance. To avoid the risk of mismatch, sharing inventories is a very common practice among multiple cooperating firms. Sari and Hennet [86] investigated the optimal ordering and collaborative inventory planning problem in a distribution network, in which the retailers can exchange their ordered products to better match local demands. The research examines the stability of the grand coalition and appropriate pricing policy to guarantee the rational allocation of the benefits. Jouida et al. [87] investigated a coalition-formation cooperative replenishment problem with a single supplier and multiple firms, and proposed an exact cooperative replenishment algorithm which can guarantee the stability of the grand coalition. The research also examined the stability conditions regarding the ordering and transportation cost structures.

Yu et al. [88] studied the problem of production/service capacity sharing for several independent firms. The research reveals that under either the first-come, first-served policy or an optimal priority policy, capacity sharing is beneficial and there exists a cost allocation which is located in the core. However, when the firms have heterogeneous work contents and service variabilities, capacity sharing may not be beneficial.

In the past several years, collaborative consumption has become very popular, such as Uber or Airbnb in ridesharing or house-renting, etc. Lu and Quadrifoglio [89] applied the cooperative game theory to formulate the ridesharing market design as a fair cost allocation problem, and used the nucleolus solution to divide the cost. Because the nucleolus is very difficult to calculate, the authors developed a coalition generation procedure to computer the nucleolus and approximate nucleolus. Such model and results also fit well in the automatic vehicle systems.

3.2.2. Vertical cooperation

Vertical Type 1

Lippman et al. [90] used the Nash bargaining approach to investigate the design of procurement contracts under cost uncertainty and showed that cost-plus contracts dominate other cost-sharing contracts.

In reality, a project consisting of several activities may not be realized according to plan. If the project is delayed, a penalty arises, otherwise, a reward arises. For such project games, Estevez-Fernandez [91] investigated the reward allocation of expedited projects and the cost allocation of delayed projects, and studied the nonemptiness of the core.

Vertical Type 2

Xiao et al. [92] investigated the cooperative advertising among one manufacturer and multiple retailers where the coalitions' characteristic values are the associated optimal expected profit. The research shows that the core of the game is nonempty, and proposes two easy-to-implement and reasonable allocation approaches to divide the total expected profit. Using the Conditional Value at Risk as the utility measure, Chen et al. [93] investigated the stable and coordinating contract of a supply chain with a single risk-averse retailer and multiple risk-averse suppliers. However, not all coordinating contracts are stable because of the competition. Thus they proposed the notion of contract core and the concept of contract equilibrium. The former is focused on the share of profit among different firms, while the latter provides conditions on how the payment changes with the order quantity.

Mohammaditabar et al. [94] investigated the selection of the suppliers with capacity constraints. The suppliers may cooperate with each other by forming a coalition or may not cooperate by acting independently. This paper uses the cooperative and non-cooperative game theory to analyze the supplier selection and total cost of the supply chain. And the research shows that the cooperative game among the suppliers and the buyer has a non-empty core, and compares the allocation based on the core concept and those based on the Shapley value.

Vendor-managed inventory (VMI) has been regarded as a form of cooperation that can reduce economic and environmental impact of transportation, and consequently improve eco-efficiency of the supply chain. Establishing viable forms of VMI cooperation requires a fair distribution of the cooperation's economic benefits. Stellingwerf et al. [95] used the Shapley value to allocate the monetary gains in a way that reflects the partners' contributions and emissions savings.

The research topic, model properties and solution concepts involved for each reviewed paper are summarized in the following Table 2.

4. Conclusion and further research directions

In this paper, we have conducted a comprehensive survey about the recent developments of the core, Shapley value, nucleolus and Nash bargaining solution, and the applications of the four solution concepts in operations management in the past decade. Combining the above literature review with our analyses, we find the following problems that may deserve further investigation.

Table 2
Stochastic Models.

Article	Topic	Model Properties		Solution Concepts			
		Single/ Multi Period	Vertical/ Horizontal	Core	Shapley Value	Nucleolus	Nash Bargaining
Anily et al. (2017)	line balancing	single	horiz.	•			
Bendel et al. (2018)	resource pooling	single	horiz.	•	•	•	
Çetiner et al. (2013)	airline alliances	single	horiz.			•	
Chen et al. [93]	coordinating contracts	single	verti.	•			
Estevez-Fernandez [91]	activities cooperation	single	verti.	•			
Hagspiel [85]	reliability	single	horiz.		•		
Hu et al. [77]	airline alliance	single	horiz.	•			•
Jouida et al. [87]	cooperative replenishment	single	verti.	•			
Karsten et al. (2014)	spare parts pooling	multi.	horiz.	•	•		
Karsten et al. [72]	resource pooling	single	horiz.	•			
Kimms et al. (2012)	airline alliance	single	horiz.	•		•	
Lai et al. [84]	cooperative inventory	single	horiz.	•	•		
Li et al. [83]	inventory game	single	horiz.	•			
Lippman et al. [90]	project management	single	verti.				•
Liu, Lu et al. [81]	rescheduling problem	single	horiz.	•	•		
Lu et al. (2019)	ridesharing service	single	verti.	•		•	
Mohammaditabar (2016)	suppliers cooperating	single	verti.	•	•		
Sari et al. (2016)	cooperative inventory	multi	horiz.	•			
Schlicher et al. [73]	resource pooling	single	horiz.	•	•		
Stellingwerf et al. [95]	VMI cooperation	single	verti.		•		
Timmer et al. [82]	inventory cooperation	multi	horiz.	•	•		
Xiao et al. [92]	cooperative advertising	single	verti.	•	•		
Yu et al. [88]	capacity sharing	single	horiz.	•	•		
Zeng et al. [76]	capactity transfer	single	horiz	•	•	•	

Note: verti.–vertical, horiz.–horizontal.

Competitive-cooperative Problem: From the above literature, we can find that most papers only consider the cooperation among the firms. Because these cooperative firms that pool their resources actually belong to the same class, the competition is inevitable. In reality, it is the cooperation and competition that coexist. Under such background, how to frame and study the competing and cooperating problem is more realistic and challenging. Moreover, little research considers the transaction cost among the firms. Even the transaction cost may reduce the firms' expected profits, the agents may still be willing to cooperate because it will help hedge the market risk. It is also very interesting to incorporate the transaction cost and the market risk.

Multi-products cooperative problem: In the above survey, we find that almost each model is only concerned with one product. In reality, it is more common that the firms may cooperate with multiple products. Under such background, how to define the core and examine its nonemptiness deserves further investigation. On the other hand, when the firms cooperate by sharing some facilities, there may exist the sharing of several kinds of facilities among the firms in reality. However, the existing works only consider the sharing of one facility. Thus, the research about the sharing of multiple facilities also deserves further investigation.

Dynamic cooperative problem: Table 1 and Table 2 also implies that almost all works consider the static cooperation problem, however, a long-term cooperation evolving with time is often necessary. Multi-period cooperation may involve a dynamic programming problem for each firm. How to establish a dynamic cooperative approach and to solve dynamic cooperative problem is very necessary and challenging.

Multi-to-multi vertical cooperative problem: Existing vertical cooperation works focus on the purely linear cooperation or the one-to-multi cooperation, there are few studies that consider the multi-to-multi vertical cooperation. In each level, there are multiple firms that compete and cooperate with other firms in other levels, which is more realistic. It is interesting and challenging to explore the multi-to-multi vertical cooperative problem.

Choice problem of allocations in core: For a cooperative game, whether the core of the game is nonempty is very important, which will determine the stability of the cooperation. From Table 1 and Table 2, we find that almost every paper investigates whether the core of the game is nonempty. Because the core usually includes many feasible allocations, some papers give rise to the last allocation based on some other considerations, and other papers do not. Then a natural question is how to give the last allocation in the core for a practical problem. Different problems may give rise to different answers, it depends on the real practical needs.

Stability problem of the Shapley value: In operations management, the Shapley value is often applied to study the cooperative problems. The Shapley value can give rise to a unique solution for the allocation problem, but it cannot guarantee the stability of the cooperation. It may be necessary to check whether the corresponding Shapley value lies in the core. Because if the Shapley value is not in the core, the cooperation may become implementable. Leng et al. [16] made a good attempt in this direction, but the research is only limited to the ADMC games. It still needs further exploration for general cooperative games. On the other hand, the marginal contributions are all based on the surplus of the expected profits. But different agents may have different utilities for the same profit. Thus, it is very interesting and challenging to incorporate the agents' risk attitude in the Shapley value's applications in operations management.

Choice problem of solution concepts: The core just discusses the existence of stable allocations and often does not give rise to a unique solution for the allocation problem. The Shapley value, the nucleolus and the Nash bargaining scheme can generally provide a unique allocation. But the Nash bargaining scheme is mostly used to analyze two players' cooperation, the Shapley value is mainly based on the players' contribution and the nucleolus is more concerned with the satisfactoriness of each player/coalition. Thus, if the players need to emphasize the contributions in allocating the grand payoff, the Shapley value may be more reasonable, however, if the organization needs to pay more atten-

tion to the satisfaction of each coalition, the allocation in terms of the nucleolus may be more appropriate. Of course, which allocation method should be chosen depends on the problems at hand.

As far as the theoretical developments are concerned, the following two research directions still deserve further investigation.

Generalized solution concepts: First, the traditional core is developed on the basis of deterministic returns. However, if nondeterministic and stochastic returns are involved, each player/coalition's risk and ambiguity attitude will play an important role which inevitably influences the cooperation. Under such background, how to develop a generalized core which can reflect the players' contentment and realistic behavior is important and challenging. On the other hand, just like the prison games, all the individuals are rational, but the group/organization is irrational. To obtain the group's overall rationality, it may be necessary to sacrifice some agents' interest to achieve the group performance. Under this background, it is necessary to consider appropriate penalties to guarantee the grand coalition which is reflected in the least core value and the least core allocation.

In recent years, some researchers extended the Shapley value to the weighted Shapley value by incorporating different weights, which may represent different roles such as the power/worth and fairness, etc. for different players. However, how to determine the weights is very important, which may depend on the problems at hand. Similarly, how to capture the agents' different power and extend the nucleolus to the weighted nucleolus deserves further investigation. Moreover, if possible, it may be more interesting to compare the allocation in terms of Shapley value and the allocation in terms of the nucleolus, which are based on different preferences. The comparison may expose that which agents in the grand coalition prefer the Shapley value and which agents prefer the nucleolus.

For the cooperation with only two players, the Nash Bargaining approach may be the best scheme. In reality, there are often multiple players that are cooperating and bargaining the interests. How to use the Nash bargaining scheme to analyze the bargaining problem with multiple players is also important. On the other hand, some research has applied the generalized Nash bargaining scheme to study some operations management problems, which can characterize the power of different players. But how to determine the weight is also very important and challenging. These research problems and their applications in operations management deserve further exploration.

Calculation of Nucleolus problem: From Table 1 and Table 2, we can find that the number of using nucleolus as a tool is relatively small compared with the core and the Shapley value, the main reason is that the calculation of nucleolus needs to solve a series of linear programmings or a large-scale linear programming, which limits the applications of nucleolus. It is interesting and valuable to find easy methods to calculate the nucleolus for some classes of cooperative games, which will help enhance the applications of nucleolus in operations management. For example, for the three-player cooperative games or ADMC games, Leng and Parlar [96] and Leng et al. (2022) found the analytical solutions of the nucleolus. On the other hand, if the core of game is nonempty, the nucleolus must be located in the core as mentioned above. However, if the core is empty, the allocation in terms of the nucleolus cannot be stable. Thus, it may still be necessary to check the nonemptiness of the core when using nucleolus in operations management.

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